

1. Create a dbt project with snowflake connector

```
(wh) annie@mac homework_8 % dbt debug
21:27:09 Running with dbt=1.11.8
21:27:09 dbt version: 1.11.8
21:27:09 python version: 3.13.5
21:27:09 python path: /Library/Frameworks/Python.framework/Versions/3.13/bin/python3
21:27:09 os info: macOS-26.3.1-arm64-arm-64bit-Mach-O
21:27:09 Using profiles dir at /Users/annie/.dbt
21:27:09 Using profiles.yml file at /Users/annie/.dbt/profiles.yml
21:27:09 Using dbt_project.yml file at /Users/annie/Library/CloudStorage/OneDrive-Personal/Academics/Masters/SJSU/DATA-226/Homeworks/Homework_8/dbt_project.yml
21:27:09 adapter type: snowflake
21:27:09 adapter version: 1.11.4
21:27:09 Configuration:
21:27:09   profiles.yml file [OK found and valid]
21:27:09   dbt_project.yml file [OK found and valid]
21:27:09 Required dependencies:
21:27:09   - git [OK found]

21:27:09 Connection:
21:27:09   account: sfedu02-eab27764
21:27:09   user: GECKO
21:27:09   database: USER_DB_GECKO
21:27:09   warehouse: GECKO_QUERY_WH
21:27:09   role: TRAINING_ROLE
21:27:09   schema: analytics
21:27:09   authenticator: None
21:27:09   oauth_client_id: None
21:27:09   query_tag: None
21:27:09   client_session_keep_alive: False
21:27:09   host: None
21:27:09   port: None
21:27:09   proxy_host: None
21:27:09   proxy_port: None
21:27:09   protocol: None
21:27:09   connect_retries: 1
21:27:09   connect_timeout: None
21:27:09   retry_on_database_errors: False
21:27:09   retry_all: False
21:27:09   insecure_mode: False
21:27:09   reuse_connections: True
21:27:09   s3_stage_vpce_dns_name: None
21:27:09   platform_detection_timeout_seconds: 0.0
21:27:09 Registered adapter: snowflake=1.11.4
21:27:10 Connection test: [OK connection ok]

21:27:10 All checks passed!
```

```
✓ homework_8
  > analyses
  > logs
  > macros
  > models/transform
  > seeds
  > snapshots
  > tests
  ◆ .gitignore
  ! dbt_project.yml
  ⓘ README.md
```

2. Set up transform models

- Input (transform) tables should be built as CTE (involves changes to

```
! dbt_project.yml ×
homework_8 > ! dbt_project.yml
32  models:
33    homework_8:
34      +materialized: view
35
36    transform:
37      +materialized: ephemeral
38
39    analytics:
40      +materialized: table
41
```

dbt_project.yml)

- Show contents of models/transform/user_session_channel.sql and transform/session_timestamp.sql

```
session_timestamp_cleaned.sql ×
homework_8 > models > transform > session_timestamp_cleaned.sql
1  select sessionID, ts
2  from {{ source('raw', 'session_timestamp') }}
3  where sessionID is not null
4
```

```
user_session_channel_cleaned.sql ●
homework_8 > models > transform > user_session_channel_cleaned.sql
1  SELECT userId, sessionId, channel
2  FROM {{ source('raw', 'user_session_channel') }}
3  WHERE sessionId IS NOT NULL
```

```

• (wh) annie@mac homework_8 % dbt run
00:35:10 Running with dbt=1.11.8
00:35:10 Registered adapter: snowflake=1.11.4
00:35:10 Unable to do partial parsing because a project config has changed
00:35:10 [WARNING]: Configuration paths exist in your dbt_project.yml file which do not apply to any resources.
There are 1 unused configuration paths:
- models.homework_8.analytics
00:35:10 Found 2 models, 2 sources, 523 macros
00:35:10
00:35:10 Concurrency: 1 threads (target='dev')
00:35:10
00:35:10 Finished running in 0 hours 0 minutes and 0.01 seconds (0.01s).
○ (wh) annie@mac homework_8 % █

```

3. Set up analytics models (1pt)

a. Show contents of models/analytics/session_summary.sql

 session_summary.sql ×

```

homework_8 > models > analytics >  session_summary.sql
1  SELECT u.userId, u.sessionId, u.channel, st.ts
2  FROM {{ ref('user_session_channel_cleaned') }} u
3  JOIN {{ ref('session_timestamp_cleaned') }} st
4  ON u.sessionId = st.sessionId

```

```

• (wh) annie@mac homework_8 % dbt run
00:40:18 Running with dbt=1.11.8
00:40:18 Registered adapter: snowflake=1.11.4
00:40:18 Found 3 models, 2 sources, 523 macros
00:40:18
00:40:18 Concurrency: 1 threads (target='dev')
00:40:18
00:40:20 1 of 1 START sql table model analytics.session_summary ..... [RUN]
00:40:24 1 of 1 OK created sql table model analytics.session_summary ..... [SUCCESS 101520 in 4.42s]
00:40:25 Finished running 1 table model in 0 hours 0 minutes and 6.44 seconds (6.44s).
00:40:25
00:40:25 Completed successfully
00:40:25
00:40:25 Done. PASS=1 WARN=0 ERROR=0 SKIP=0 NO-OP=0 TOTAL=1
○ (wh) annie@mac homework_8 % █

```

4. Add Snapshot to the analytics table (session_summary)
 - a. Show contents of snapshots/snapshot_session_summary.sql

```
snapshot_session_summary.sql ×

homework_8 > snapshots > snapshot_session_summary.sql
1  {% snapshot snapshot_session_summary %}
2
3  {{
4      config(
5          target_schema='snapshot',
6          unique_key='sessionId',
7          strategy='timestamp',
8          updated_at='ts',
9          invalidate_hard_deletes=True
10     )
11 }}
12
13 SELECT * FROM {{ ref('session_summary') }}
14
15 {% endsnapshot %}
```

```
(wh) annie@mac homework_8 % dbt run
00:44:43 Running with dbt=1.11.8
00:44:44 Registered adapter: snowflake=1.11.4
00:44:44 Found 3 models, 1 snapshot, 2 sources, 523 macros
00:44:44 Concurrency: 1 threads (target='dev')
00:44:44
00:44:50 1 of 1 START sql table model analytics.session_summary ..... [RUN]
00:44:52 1 of 1 OK created sql table model analytics.session_summary ..... [SUCCESS 101520 in 2.03s]
00:44:52 Finished running 1 table model in 0 hours 0 minutes and 7.82 seconds (7.82s).
00:44:52 Completed successfully
00:44:52 Done. PASS=1 WARN=0 ERROR=0 SKIP=0 NO-OP=0 TOTAL=1
```

5. Add at least 2 tests to sessionId field of session_summary
 - a. Show contents of models/schema.yml

```
! schema.yml ×
homework_8 > models > ! schema.yml
1  version: 2
2
3  models:
4    - name: session_summary
5      description: "Analytics model for session data"
6      columns:
7        - name: sessionId
8          description: "Unique identifier for each session"
9          tests:
10            - unique
11            - not_null
```

```
● (wh) annie@mac homework_8 % dbt test
01:13:00 Running with dbt=1.11.8
01:13:00 Registered adapter: snowflake=1.11.4
01:13:01 Found 3 models, 1 snapshot, 2 data tests, 2 sources, 523 macros
01:13:01 Concurrency: 1 threads (target='dev')
01:13:01
01:13:03 1 of 2 START test not_null_session_summary_sessionId ..... [RUN]
01:13:03 1 of 2 PASS not_null_session_summary_sessionId ..... [PASS in 0.47s]
01:13:03 2 of 2 START test unique_session_summary_sessionId ..... [RUN]
01:13:04 2 of 2 PASS unique_session_summary_sessionId ..... [PASS in 0.63s]
01:13:04
01:13:04 Finished running 2 data tests in 0 hours 0 minutes and 3.01 seconds (3.01s).
01:13:04 Completed successfully
01:13:04
01:13:04 Done. PASS=2 WARN=0 ERROR=0 SKIP=0 NO-OP=0 TOTAL=2
○ (wh) annie@mac homework_8 %
```